

A Validation Study of the BRIEF-2 Among Kindergarteners and First Graders At-Risk for Behavior Problems

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Abstract

Early identification of executive dysfunction and timely school-based intervention efforts are critical for students at risk for problematic behaviors during early elementary school. The original Behavior Rating Inventory of Executive Functioning (BRIEF) was designed to measure real-world behavioral manifestations of executive functioning, neurocognitive processes critical for school success. With the updated BRIEF-2, independent validation is needed with kindergarten and first grade students at risk for emotional and behavioral disorders. Thus, using item level analyses, we examined the factor structure of the BRIEF-2 Teacher Rating form with 1,112 students. Results indicated little evidence for the original three-index model and supported a modified two-index model, with a Cognitive Regulation Index and an overall Behavior–Emotion Regulation Index. Criterion related validity indicated positive relationships with performance-based executive functioning (Head-Toes-Knees-Shoulders) and later internalizing and externalizing behaviors. We discuss implications of findings for early identification and school-based intervention efforts, as well as future research.

Keywords

BRIEF, executive function, factor analysis, validity, at-risk students

The transition to elementary school presents a period of vulnerability for children at risk of significant emotional and behavioral difficulties. Approximately one in three children experience mental health problems (Merikangas et al., 2009), yet compared with adolescents, young children with early onset emotional and behavioral difficulties are more likely to experience low academic achievement, school dropout, peer rejection, incarceration, and unemployment in later years (Bradley et al., 2008; Kauffman & Landrum, 2017; Morgan et al., 2009; Ruchkin et al., 2003). Despite the importance of identification and intervention to mitigate adverse outcomes (Poulou, 2015), less than 1% of the 56 million students in the United States receive individualized school-based services for mental health (i.e., under the special education classification of emotional and behavioral disorders; Kauffman & Landrum, 2017; Office of Special Education Programs, 2017). Of these students, many receive school-based support much later than students with other types of disabilities (Wagner et al., 2005). This is particularly concerning given that chronic behavioral problems are less amenable to intervention as students progress through elementary and middle school (Bradley et al., 2008; Yeager et al., 2017).

Fortunately, as a pivotal intervention period (Bradley et al., 2008; Milkie & Warner, 2011; Ruchkin et al., 2003),

early elementary school provides a leverage point for timely identification, especially during kindergarten and first grade when students are experiencing rapid brain maturation (Best & Miller, 2010; Zelazo & Carlson, 2012) and developing foundational skills (Blair & Diamond, 2008; Hamre & Pianta, 2001). Yet, timely and accurate identification is dependent on the use of reliable and valid measures that assess factors underlying psychological functioning (Goldstein & Flake, 2016). One such factor is executive function (EF)—a set of neurocognitive processes that underlie goal directed behaviors (Miyake et al., 2000; Zelazo et al., 2016).

Multiple theories have shaped understanding of EF, with conceptualization and measures of EF varying within and across disciplines (see Nigg, 2017). Yet there is general agreement that EF is an overarching construct for top-down,

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interrelated attention-regulation processes coordinated by the prefrontal cortex that underlie goal directed actions (Miyake et al., 2000; Zelazo et al. 2016), with some definitions emphasizing rule-governed cognition (Nigg, 2017). There is variability, however, in terms of which distinct components make up EF. Increasingly, scholars posit that EF comprises three unique yet related processes of (a) working memory, which is responsible for temporarily storing, retaining, and recalling information; (b) inhibitory control, the ability to control a habitual or proponent response; and (c) cognitive flexibility, the capacity to shift mental sets or adapt to task or contextual demands (Best & Miller, 2010; Miyake et al., 2000; Zelazo et al., 2016). These, in turn, support higher order skills, such as self-regulation and problem solving (Diamond, 2013; Nigg, 2017). Other commonly identified EF components include interference control, fluency, planning, and reasoning (Diamond, 2013; Nigg, 2017). Overall, EF processes enable individuals to pay attention, stop impulses, ignore distractions, remember and use information, and adapt to situations, which are essential skills for school and life.

Over two decades of research has highlighted the central role of EF in children's cognitive, behavioral, and psychological development (Diamond, 2013; Zelazo, 2020). EF is an established predictor of life-long achievement, social skills, and healthy interpersonal relationships (Ahmed et al., 2018; Holmes et al., 2016; Moffitt et al., 2011; Pascual et al., 2019). It is often a better indicator of children's school readiness and performance, including behavioral competence (Morgan et al., 2009, 2019), than previous academic knowledge or IQ (e.g., Duckworth & Seligman 2005; McClelland et al., 2007). In contrast, EF deficits are associated with an array of maladaptive behaviors (see meta-analysis, Schoemaker et al., 2013) and mental health problems (Zelazo, 2020), including attention deficit hyperactivity disorder (ADHD), oppositional defiant disorder, conduct disorder, and other emotional and behavioral disorders (e.g., Cumming et al., 2019; see meta-analysis Noordermeer et al., 2016). Thus, when identifying students at risk for behavior and mental health problems in schools and developing targeted services (Wass et al., 2012; Willoughby et al., 2011; Willoughby et al., 2016), school-based professionals (e.g., school psychologists) should use valid and reliable measures of EF.

The most widely used EF measure (Roth et al., 2014) is the Behavior Rating Inventory of Executive Function (BRIEF; Gioia et al., 2015), which is an ecological-based measure intended to capture real-world *behavioral* manifestations of EF dysfunction. Isquith et al. (2013) posit that rating scales, such as the BRIEF, are not limited to assessing specific EF components in isolation (e.g., working memory), but rather assess use of these skills in real life contexts. The original rating scale was validated across multiple age groups and is well known to be sensitive in identifying clinically significant behaviors (e.g., ADHD, autism, traumatic

brain injury; Isquith et al., 2013; Roth et al., 2014). Yet with the development of a new version of the Behavior Rating Inventory of Executive Function—Second Edition (BRIEF-2; Gioia et al., 2015), few studies have examined its validity with early elementary students at-risk for behavior problems. Therefore, the purpose of this article is to (a) highlight commonly used methods of measuring childhood EF, (b) review what is currently known about the BRIEF-2's validity and reliability, and (c) present findings and implications from our validation study of the BRIEF-2 with kindergarten and first grade students who are at risk for emotional and behavioral disorders.

Measuring Executive Functioning to Identify At-Risk Children

Current EF measures are based largely on EF theoretical models (e.g., Miyake et al., 2000), with scholars assessing skills through either performance based or rating scale instruments. Researchers have tended to view performance-based EF tasks (e.g., Head-Toes-Knees-Shoulders) that are administered directly to children under standardized conditions as the gold standard for measuring EF (Zelazo et al., 2013). Yet, because these tasks tap into limited components of EF for a short time, some scholars argue that they do not capture the multidimensional processes or behaviors often demonstrated in real-world situations (Fuhs et al., 2015). This draws into question their ecological validity and generalizability to classroom settings (Toplak et al., 2013).

In contrast, rating scales or questionnaires gather multiple informants' (e.g., parents, teachers) perceptions of children's behavioral and affective manifestations of EF (e.g., emotional regulation) across home and school settings (Gioia et al., 2000). Because rating scales, such as the BRIEF (Gioia et al., 2000), are ecologically valid, time and cost-efficient, and sensitive to identifying individuals with or at risk for emotional and behavioral difficulties (Isquith et al., 2014; Miranda et al., 2015), school-based professionals and practitioners often rely on them. Yet studies have found weak relationships between performance based and EF ratings scales with samples of children with psychiatric conditions and developmental disorders (Isquith et al., 2014; McAuley et al., 2010; Toplak et al., 2013; Vriezen & Pigott, 2002). Scholars posit weak correlations are likely because they assess different underlying constructs, such that performance-based tasks tend to capture "cool" cognitive EF (e.g., inhibitory control, working memory, cognitive flexibility), whereas rating scales likely assess the application of "contextualized" EF skills in everyday settings (Isquith et al., 2014; Zelazo & Carlson, 2012). Although additional research is needed to determine which specific constructs each measures, Isquith et al. (2014) recommend the use of both performance-based assessments and rating scales to accurately capture students' EF strengths and weaknesses.

To be reliable and valid, EF measures must not only be able to distinguish between typical and atypical development but also sensitive to age differences (Best & Miller, 2010). EF develops throughout childhood and early adulthood, with active maturation periods in preschool and middle school (Best & Miller, 2010; Diamond, 2013). Certain EF processes are posited to develop first, such as inhibitory control, which shows significant improvement during preschool (Best & Miller, 2010). In accordance with these developmental changes in EF (Best & Miller, 2010), researchers have established age-corrected scores, such as the performance-based NIH Toolbox (Weintraub et al., 2013) or separate measures based on age, such as the BRIEF-Preschool Version (Skogan et al., 2016). Yet there is currently limited independent factor analytic studies of the BRIEF-2 (Muñoz & Filippetti, 2019) that examine targeted age groups.

Development of the BRIEF-2

The BRIEF-2 (Gioia et al., 2015) was recently developed to address some of the limitations of the BRIEF (Gioia et al., 2000). The original 86-item, BRIEF is a parent- and teacher-completed questionnaire that consists of eight scales (Inhibit, Shift, Emotional Control, Initiate, Working Memory, Plan/Organize, Organization of Materials, and Monitor) forming two broad indexes: Behavior Regulation Index (BRI; Inhibit, Shift, and Emotional Control scales) and Metacognition Index (MI; Initiate, Working Memory, Plan/Organize, Organization of Materials, and Monitor scales). Although the BRIEF manual provides support for a two-factor model made up of BRI and MI, several studies have found subscale cross loadings and proposed factor structure inconsistency. For instance, the BRIEF developers found that Inhibit loaded on both BRI and MI (Gioia et al., 2000), but opted to base loading decisions on theoretical considerations. Other researchers have also found inconsistency in scale loadings. For instance, Donders et al. (2010) noted Inhibit had significant cross loadings on both MI and BRI (assigned index in the normed version). Studies with clinical samples found Monitor not only loaded onto both indexes but also discovered Monitor was likely better represented by two scales made up of monitoring tasks and monitoring self (e.g., Gioia et al., 2002; Lyons Usher et al., 2016). Furthermore, several confirmatory factor analysis (CFA) studies of both the BRIEF Parent and Teacher forms found stronger support for a three-factor model instead of a two-factor model for both typically developing and clinical groups (Egeland & Fallmyr, 2010; Gioia et al., 2002; Peters et al., 2012). Thus, several researchers have raised questions about the internal validity of the original version of the BRIEF (e.g., Peters et al., 2012).

With the goal to “enhance its utility for clinical and research purposes without substantially altering the scales”

(p. 62), Gioia et al. (2015) created the revised BRIEF-2. The developers reduced the number of items, added a new Emotional Regulation Index (ERI), renamed MI as the Cognitive Regulation Index (CRI), and separated the Monitor scale into two scales (Task-Monitor and Self-Monitor). The revised version consists of 63 items (rated 1 = never, 2 = sometimes, 3 = always) that includes nine clinical scales (Inhibit, Self-Monitor, Shift, Emotional Control, Initiate, Working Memory, Plan/Organize, Task-Monitor, and Organization of Materials) and three composite indexes: BRI (Inhibit and Self-Monitor scales), ERI (Emotional Control and Shift scales) and CRI (Initiate, Working Memory, Plan/Organize, Task-Monitor, and Organization of Materials scales). When standardizing the BRIEF-2 Teacher Form ($N = 1,400$) among a sample of students aged 5 to 18 years ($M = 11.41$, $SD = 4.00$), Gioia et al. (2015) found items adequately loaded on all nine scales, and a three-factor (index) model demonstrated good fit, ($\chi^2 [1,400] = 415.38$, $p < .001$ comparative fit index [CFI] = .97, standardized root mean square residual [SRMR] = .03, root mean square error of approximation [RMSEA] = .12). Furthermore, each index demonstrated strong internal reliability, with coefficients of .95 (BRI), .94 (ERI), and .98 (CRI). The developers normed the BRIEF-2 by gender and for the following age groups: 5 to 7 years, 8 to 10 years, 11 to 13 years, and 14 to 18 years.

Limitations, however, have been noted in previous BRIEF-2 validation studies. First, much of the previous analyses has been conducted using parceling at the scale level (Gioia et al., 2015) versus at the item level, which not only makes the assumption that theoretically assigned items accurately reflect the assigned scale but may also mask item-level sources of misfit (Little et al., 2013), leading to biased estimates (Bandalos & Finney, 2001; Jacobson et al., 2019). Second, several choices related to underlying scales have been made based on theoretical considerations versus empirical methods (e.g., Gioia et al., 2015). As highlighted by Jacobson et al. (2017), although theory provides a beneficial starting point, the validity of an instrument must be based on empirical evidence, which, in turn, can further shape theory. Overall, for the BRIEF-2 to have clinical utility, additional validation studies are necessary.

Purpose

To our knowledge, there has been no independent examination of the factor structure of BRIEF-2 Teacher Form with early elementary school-age students with or at risk for emotional and behavioral disorders, which is critical due to notable developmental changes in EF and risk for early onset behavior problems. Therefore, to accurately identify executive dysfunction in this age group, the purpose of our study was to examine the internal factor structure, criterion-related validity, and measurement reliability of the BRIEF-2

Teacher Form at the item level using an iterative process that included both CFA and exploratory factor analysis (EFA). We hypothesized that the developer specified factor structure of the BRIEF-2 would not provide adequate fit to the data, due to (a) previously noted misalignment between theoretical derivation and empirical evidence underlying the BRIEF-2, and (b) significant developmental differences in EF for this age group. We anticipated that a more parsimonious model would not only demonstrate good internal validity and reliability but also criterion-related validity (e.g., performance-based EF, behavioral ratings).

Method

Sample

We used data from a 3-year cluster randomized controlled trial investigating the efficacy of the Social Emotional Learning Foundations (SELF, see Daunic et al., 2013) program, which is a targeted (i.e., Tier 2) literacy-integrated social-emotional curriculum for kindergarten and first grade students who are at risk for developing emotional and behavioral disorders. To identify students, teachers used Stages 1 and 2 of the Systematic Screening for Behavior Disorders (Walker & Severson, 1992), which involves a structured process for screening children at risk for behavioral or emotional difficulties. In Stage 1, teachers first read research-validated definitions of both internalizing (e.g., depression) and externalizing (e.g., aggression) behavioral dimensions. Then they selected four to five students for each dimension based on their evaluations of student's behavioral characteristics. In Stage 2, teachers completed the critical events index and the combined frequency index, which identified the top two to three students on each dimension for possible participation in the study. The SSBD has strong psychometric properties including internal consistency ($\alpha = .84-.86$) test-retest reliability ($\rho = .74 - .88$), interrater reliability ($\rho = .72 - .94$), and convergent validity (Glover & Albers, 2007; Lane et al., 2014).

A total of 304 teachers (163 kindergarten and 141 first grade) and 1,154 students (627 kindergarteners and 527 first graders) participated in the study, of which data from 1,112 participants was used in our analyses (due to missing data). Schools were randomly assigned to treatment or control. Demographic data were available for 94% of students due to some districts not providing demographic data. The majority of students were male (62%), 64% identified as White, 20% as African American, 9% as Hispanic, and 7% as Other. Approximately, half of the students were in kindergarten (54%), 3% were English learners, and 20% received special education services. Yet students were excluded if they were identified with a developmental delay to eliminate a potentially confounding variable in determining treatment efficacy (i.e., related to the larger intervention

study). Participating students came from one southeastern state and from 52 schools in which a majority (81%) of the student population were eligible for free or reduced lunch.

We used the full sample ($N = 1,112$) for primary analysis and then split the data into two equal subsamples to investigate potential model structure (first half) and conduct cross-validation (second half), as explained in the analysis section. For the subsample used to investigate model structure, 56% were in kindergarten, 64% were male, 56% identified as White, 27% as African American, 11% as Hispanic, and 6% as Other. Approximately 86% received free or reduced lunch, 5.6% were English learners, and 21% received special education services. For the subsample used for cross-validation, 52% were in kindergarten, 72% of the sample identified as White, 13% as African American, 7% as Hispanic, and 8% as Other. The majority received free or reduced lunch (75%), 1% were English learners, and 20% received special education services. We conducted chi-square tests to examine whether systematic differences existed between subsamples and found no statistically significant differences in terms of gender, grade level, and BRIEF-2 item responses.

Additional Measures

In addition to BRIEF-2 raw data, the research team assessed students on a performance-based measure of EF and collected teacher reports of externalizing and internalizing behaviors at both preintervention and postintervention. Given the link between EF and behavioral functioning (Riggs et al., 2004; Schoemaker et al., 2012; White et al., 2013), we used data from these measures at preintervention (control and treatment groups) and postintervention (control group only) to assess the convergent and predictive validity of BRIEF-2.

Head-Toes-Knees-Shoulders Task. To assess EF, we used raw scores from the Head-Toes-Knees-Shoulders task (HTKS; Ponitz et al., 2009), which is a direct assessment of overall EF (i.e., working memory, cognitive flexibility, and inhibitory control) for students aged 4 to 6. In a game-like format, across 10 trials, children are first instructed to respond naturally to the directions of "touch your head" and "touch your toes" followed by directions to respond in the opposite fashion (i.e., touch toes when asked to touch head). In the second part of the assessment, children are instructed to touch their knees and shoulders and then to do the opposite; students are then prompted with each of the four commands (e.g., head, toes, knees, or shoulders) and instructed to respond in the opposite for the next 10 trials. Students receive scores of 0 (incorrect), 1 (self-correct), or 2 (correct). Based on six practice opportunities (three for each part) and 20 trials, scores range from 0 to 52 with higher scores representing higher EF abilities. In the current sample, the HTKS demonstrated

strong internal consistency ($\alpha = .95$). Previous psychometric studies of the HTKS have established construct validity, predictive validity, test–retest reliability ($r = .74, p < .001$), and interrater reliability ($\kappa = .79, p < .001$; McClelland et al., 2014; Ponitz et al., 2009).

Clinical Assessment of Behavior Teacher Rating Form (CAB-T). We used raw scores from the CAB-T (Bracken & Keith, 2004) to measure student behavioral functioning and school adjustment. Comprised of two behavioral scales (Internalizing and Externalizing) and two adaptive scales (Social Skills and Competence), teachers answer 70 questions using a Likert-type format from 1 (*always or very frequently*) to 5 (*never*) with lower scores representing increased problems. For the present study, we used only the scores of the Internalizing (e.g., “is sad for long periods of time”) and Externalizing (e.g., “is defiant of people in authority”) scales to assess the relationship between BRIEF-2 scales and emotional and behavioral functioning. In the current sample, Cronbach’s alphas for the Internalizing scale were .88 at pre and .90 at post; the Externalizing scale was .97 at pre and .97 at post. Psychometrics of the CAB-T have been established related to content validity, construct validity, and convergent validity (Bracken & Keith, 2004).

Analyses

Internal Factor Structure. Following similar analytic procedures as Jacobson et al. (2019), who investigated whether EF theory underlying the BRIEF-2 Parent Form was empirically supported using item level data, we examined the BRIEF-2 Teacher Form internal factor structure using an iterative process. First, with the full sample ($N = 1,112$; deidentified data and codes are available on request from the corresponding author), we evaluated the BRIEF-2 internal factor structure by fitting the a priori manual specified three-index model (referred to as manual model hereinafter; Gioia et al., 2015; see Figure 1), yet examined items at the item level, such that 60 items loaded onto 9 scales (first order factors)—Inhibit (eight items), Self-Monitor (five items), Shift (eight items), Emotional Control (eight items), Initiate (four items), Working Memory (eight items), Plan/Organize (eight items), Task Monitor (six items), Organization of Materials (five items). Three additional items made up an Infrequency scale, which provided insight into whether respondents completed the measure in an atypical manner. In turn, BRI (Inhibit, Self-Monitor), ERI (Shift, Emotional Control), and CRI (Initiate, Working Memory, Plan/Organize, Task-Monitor, Organization of Materials) made up three indexes (second-order factors; see Figure 1). For all first order factors, we fixed the loadings of the first item to 1 and the variance of the second-order factors to 1 and mean to 0. Second, due to past support for a bifactor

model of the BRIEF-2 (e.g., Pérez-Salas et al., 2016), we examined whether theoretically proposed indexes and a general overall EF factor better represented the underlying factor structure of the BRIEF-2 at the item level.

Then, due the inadmissible solutions from the manual model and bifactor model (i.e., untrustworthy model results), we examined the factor structure of the BRIEF-2 Teacher Form using both EFA and CFA. We split the data into two equal subsamples and followed a three-step process to investigate potential model structure. In Step 1, we conducted EFA on the first half of the data ($n = 556$). We used Geomin rotation method to produce large loadings on one factor, examined eigenvalues values, evaluated goodness of fit indices to determine which number of factors adequately fit the data, and identified items to exclude due to cross or weak loadings (Kline, 2016). In Step 2, based on findings from initial EFA analysis, we fit two adjusted three-factor and two-factor models using EFA. Our goal was to determine if the removal of items based on Step 1 findings would result in a three-index model that conceptually aligned with the manual model or whether a two-factor solution continued to be preferable. In Step 3, informed by Step 2 findings, we interpreted the viability of an adjusted second-order CFA model based on both theory and data with the remaining subsample ($n = 556$). Model results from this step were used to calculate Omega coefficients to determine the composite reliability of BRIEF-2 scales and indexes (Brunner et al., 2012).

For all EFA and CFA analysis, we used *Mplus* version 8 (Muthén & Muthén, 1998–2017). Due to the categorical nature of BRIEF-2 response data, we used the weighted least squares estimator with mean and variance adjustment to the chi-square statistic. This chosen estimator makes no assumption about the distribution of observed variables and is designed for ordered categorical data (Li, 2016). We used the TYPE = COMPLEX option to account for students being nested within classrooms. We used standard fit indices to assess model fit ($RMSEA < .10$; CFI and Tucker–Lewis index [TLI] $> .90$; and the SRMR $< .08$; Kline, 2016).

Criterion-Related Validity. Finally, similar to previous psychometric evaluations of the BRIEF-2 (Gioia et al., 2015; Spiegel et al., 2017), we evaluated criterion-related validity in two ways. First, we computed bivariate correlations to assess the strength of the relationship between scores on all of the BRIEF-2 first- and second-order factors (based on our revised model) and teacher reports of externalizing and internalizing behaviors and an EF performance-based measure. Second, we ran hierarchical multiple regression models to assess the unique variance explained by each of the BRIEF-2 indexes (second-order factors) in scores on our criterion-related variables. Specifically, for convergent validity, we computed bivariate correlations between preintervention scores on each BRIEF-2 factor and the HTKS

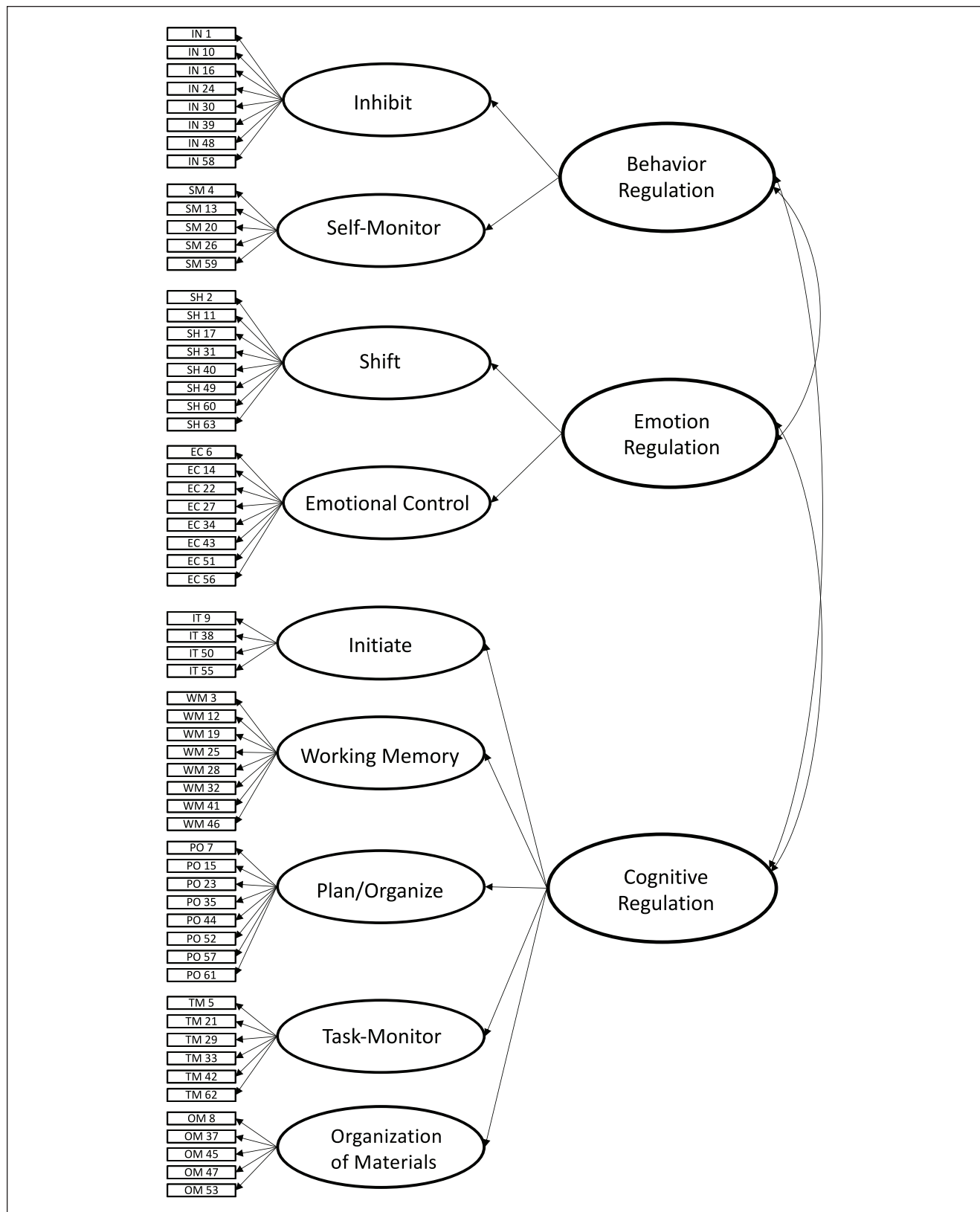


Figure 1. BRIEF-2 manual model with item level indicators.

Note. IN = Inhibit; SM = Self-Monitor; EC = Emotional Control; SH = Shift; IT = Initiate; WM = Working Memory; PO = Plan/Organize; TM = Task-Monitor; OM = Organization of Materials.

Table 1. Descriptive Statistics of BRIEF-2 Scale Scores.

BRIEF-2 Scale	Total, N = 1,112		Male, n = 654		Female, n = 396		Kindergarten, n = 604		First grade, n = 508	
	M	SD	M	SD	M	SD	M	SD	M	SD
Inhibit	16.38	5.54	17.46	5.35	14.68	5.45	16.28	5.74	16.50	5.30
Self-Monitor	10.74	3.28	11.24	3.14	9.94	3.38	10.59	3.44	10.93	3.09
Shift	15.28	4.06	15.69	3.88	14.69	4.24	15.20	4.14	15.38	3.98
Emotional Control	16.53	6.02	16.99	5.97	15.99	6.12	16.09	6.03	17.05	5.98
Initiate	8.32	2.68	8.62	2.64	7.83	2.71	8.24	2.68	8.42	2.69
Working Memory	16.63	4.96	17.23	4.68	15.66	5.22	16.47	5.04	16.83	4.87
Plan	15.62	4.45	16.14	4.28	14.84	4.63	15.34	4.55	15.94	4.32
Task-Monitor	12.88	3.96	13.58	3.75	11.76	4.00	12.54	4.10	13.30	3.75
Organization of Materials	9.08	3.18	9.50	3.11	8.40	3.17	8.85	3.16	9.35	3.18

Note. Cases with missingness on gender were not included. BRIEF-2 = Behavior Rating Inventory of Executive Function–Second edition.

and the Internalizing and Externalizing scales of the CAB-T. Next, we ran a hierarchical multiple regression model using preintervention scores on the BRIEF-2 indexes as predictors of preintervention CAB-T and HTKS scores. To evaluate predictive validity, we correlated preintervention scores on each BRIEF-2 factor with postintervention scores on the CAB-T Internalizing and Externalizing scales. We then ran a hierarchical multiple regression using preintervention scores on the BRIEF-2 indexes as predictors of postintervention scores on both CAB-T behavior scales. Postintervention scores were gathered 6 to 7 months after preintervention scores, which provides an indication of the degree to which scores on the BRIEF-2 forecasted teacher-reported maladaptive internalizing and externalizing behaviors (CAB-T).

To account for the nesting of students in classrooms (i.e., group effect), we group-mean centered (Enders & Tofighi, 2007) scores on the BRIEF-2 prior to calculating all bivariate correlations. In our regression models, we controlled for any variance explained by group effects by adding the classroom grouping variable in the first step and BRIEF-2 indexes in the second step. To eliminate treatment effects, we used preintervention scores only for convergent validity and analyzed only control group data for predictive validity. To handle missing data, we used pairwise deletion. For convergent validity analyses, the resulting samples ranged from 1,093 to 1,112 and for predictive validity analyses, the samples ranged from 476 to 526. We used SPSS v27 for all criterion-related analyses.

Results

Descriptive Statistics

We provide descriptive statistics of the mean and standard deviation of all BRIEF-2 scale scores for the sample

($N = 1,112$), gender, and grade level in Table 1. We also outline item response usage (Never, Sometimes, and Often) based on gender and grade level. See Table 1.

Internal Factor Structure

Preliminary Investigation: Manual Model and Bifactor Model. We first examined the 63-item, a priori manual specified three index model (Gioia et al., 2015) with the entire sample ($N = 1,112$), yet included item indicators for first order factors (i.e., scales). Fit indices denoted adequate model fit (RMSEA = .07, CFI = .91, TLI = .91, SRMR = .09); however, we noted the model resulted in inadmissible parameter estimates (i.e., Heywood case; Chen et al., 2001)—highlighting potential model misspecification. Based on past research supporting a bifactor model for the BRIEF (Pérez-Salas et al., 2016), we also examined a bifactor model in which items loaded both onto BRI, ERI, and CRI and an overall general EF factor. Though we found model fit improved (RMSEA = .06, CFI = .93, TLI = .93, SRMR = .08), the model also resulted in a Heywood case error. Although unexpected, previous analyses of the BRIEF have resulted in Heywood errors (e.g., Bonillo et al., 2012; Isquith et al., 2004). See Table 2 for model fit statistics for all preliminary models.

Step 1 Iterative Investigation: EFA Models. Due to inadmissible parameter estimates, we conducted an EFA ($n = 556$) to identify problematic items (e.g., weak loadings, cross-loadings) and determine whether a new factor structure could correct potential previous model misspecifications. In the initial analyses with 60 items (without infrequency scale Items 18, 36, 54), eigenvalues greater than one supported one- to seven-factor models (see Table 3). With the exception of a one-factor model that fit the data poorly (RMSEA = .10, CFI = .84, TLI = .84, SRMR = .18), the remaining

Table 2. Summary of Fit Statistics for BRIEF-2 CFA Models.

Model	χ^2 (df)	RMSEA	CFI	TLI	SRMR
Manual model ^a	11839.43 (1698)***	.07	.91	.91	.09
Bifactor model ^a	10810.86 (1650)***	.06	.93	.93	.08
Adjusted two-factor second-order CFA	3857.87 (1316)***	.06	.95	.95	.08
Final CFA model	3434.49 (1315)***	.05	.96	.96	.07

Note. BRIEF-2 = Behavior Rating Inventory of Executive Function–Second edition; *df* = degrees of freedom; RMSEA = root mean square error of approximation; CFI = comparative fit index; TLI = Tucker–Lewis index; SRMR = standardized root mean square residual; CFA = confirmatory factor analysis.

^aIndicates error related to inadmissible parameter estimates (Heywood Cases).

****p* < .001.

Table 3. Summary of Eigenvalues and Fit Statistics for BRIEF-2 EFA Models.

Model	Eigenvalue	RMSEA	CFI	TLI	SRMR
One-factor	31.14	.10	.84	.84	.18
Two-factor	8.36	.06	.93	.93	.08
Three-factor	3.71	.05	.97	.96	.04
Four-factor	1.33	.04	.97	.97	.04
Five-factor	1.22	.04	.98	.98	.03
Six-factor	1.14	.03	.99	.98	.03
Seven-factor	1.04	.03	.99	.99	.03
Eight-factor	0.89	.03	.99	.99	.02
Nine-factor	0.83	.02	.99	.99	.02

Note. BRIEF-2 = Behavior Rating Inventory of Executive Function–Second edition; RMSEA = root mean square error of approximation; CFI = comparative fit index; TLI = Tucker–Lewis index; SRMR = standardized root mean square residual.

models demonstrated adequate fit (see Table 3 for fit and eigenvalues). In EFA, it is typical for more than one model to fit the data; thus, model selection must be based on both data as well as theoretical and conceptual relevancy (Kline, 2016). When interpreting model viability, we considered items that loaded above .40 (Kline, 2016), and determined whether conceptually aligned items (e.g., BRI) loaded together. On examining item loadings, we found that the three- through seven-factor models were conceptually and analytically weak. For example, both the six- and seven-factor models had one factor with item loadings below the .40 cutoff. In the five-factor model, two factors had only two item loadings above .40, with each of these loading more strongly on other factors in the model. In the four-factor model, one factor had only three items that loaded above .40, each which cross loaded on another factor. Finally, in the three-factor model, which would have aligned with the a priori specified BRIEF-2 manual model indexes, items that conceptually would make up the CRI index tended to load together under Factor 1 and those items that aligned with the BRI index tended to load onto Factor 2. Yet the remaining items that conceptually would fall under

Table 4. Standardized Loadings for Excluded Items in the EFA Two-Factor Model.

Excluded items	Factor 1	Factor 2
INI	.35*	.48*
SH11	.22*	.48*
PO15	.36*	.42*
SH17	.39*	.39*
IN30	.36*	.53*
OM37	.51*	.33*
SH40	.34*	.44*

Note. EFA = exploratory factor analysis; IN = Inhibit; SH = Shift; PO = Plan/Organize; OM = Organization of Materials.

**p* < .05.

Factor 3 (i.e., ERI) had cross loadings with BRI, indicating that these items likely fall under one overall BRI-ERI factor. Thus, a three-factor solution was not viable.

Conversely, the two-factor model not only fit the data adequately (RMSEA = .07, CFI = .93, TLI = .93, and SRMR = .08) but also had items that loaded onto factors that conceptually aligned. For instance, theoretically aligned CRI items loaded together on Factor 1, and the majority of the BRI or ERI items loaded together on Factor 2. Yet, as demonstrated in Table 4, seven items demonstrated either low loadings (e.g., below .40) or cross loaded onto more than one factor, indicating that the item either (a) did not serve as a strong indicator of the construct, or (b) had possible overlap with multiple constructs. For example, Item 17 not only had a loading below .40 but also loaded onto both Factor 1 (.39) and Factor 2 (.39), indicating the item likely did not serve as a strong indicator of either construct. Thus, we excluded Items 1, 11, 15, 17, 30, 37, and 40.

Step 2 Iterative Investigation: Adjusted EFA Models. With the remaining 53 items, we investigated a two- and three-factor solution to determine if the removal of these items would result in a three-factor model that conceptually aligned with the manual model or whether a two-factor solution continued to be superior. A three-factor model demonstrated good

fit (RMSEA = .05, CFI = .97, TLI = .97, and SRMR = .04) and all items loaded above .40. However, considering the presence of multiple item cross loadings, we determined that a three-factor model was unsupported.

A second-factor solution not only fit the data adequately (RMSEA = .07, CFI = .94, TLI = .94, and SRMR = .07), but resulted in distinct CRI (Factor 1) and Behavior–Emotion Regulation Index (B-ERI; Factor 2; combined BRI and ERI) factors, with item loadings above .50. There is theoretical and empirical support for the overlap between BRI and ERI, such that effective behavior regulation is dependent on emotion regulation skills (Garner, 2010; Gross, 2002), and the distinction between the two for younger students may be less notable due to less developed cognitive and language abilities to express and regulate emotions and resulting behaviors (Diaz et al., 2017; Gross, 2002). Furthermore, in the original BRIEF Teacher Form, the Inhibit, Shift, and Emotional Control scales made up an overall BRI that included emotional control (a component of emotion regulation), further highlighting theoretical overlap. However, we found several items that were purported to align with B-ERI loaded more strongly on CRI. This involves Items 13, 60, and 63 that presented strong positive loadings (.66, .68, and .70, respectively) on CRI. As these items are phrased in ways that capture more cognitive processes (i.e., “thinking,” “understanding”), they may be more reflective of cognitive regulation process (e.g., task monitor, planning, and organizing) for this age group.

Step 3 Iterative Investigation: Final CFA Model. Based on EFA findings, we used the remaining subsample ($n = 556$) to test a 53-item second order CFA model made up of nine first-order factors (scales) and two distinct second-order factors—CRI (Initiate, Working Memory, Plan/Organize, Task-Monitor, Organization of Materials) and B-ERI (Inhibit, Self-Monitor, Shift, Emotional Control). See Figure 2. Item 13 loaded on Task Monitor and Items 60 and 63 on Plan/Organize. The adjusted model presented acceptable fit (RMSEA = .06, CFI = .95, TLI = .95, and SRMR = .08). Following the manual model, where residual correlations were permitted between Inhibit and Emotional Control, Working Memory, and Organization of Materials, we examined model modification indices to determine theoretically meaningful residual correlations at item level or first-order factors (i.e., BRIEF-2 scales). These help to signify potential model respecifications/improvement (Kenny, 2020). According to the results, we added the correlation of disturbances between Emotional Control and Shift, *modification index* (1) = 521.07, $p < .05$. This respecified model presented improved model fit, such that RMSEA = .05, CFI = .96, TLI = .96, and SRMR = .07. It is worth noting that the Self-Monitor scale presented a small negative residual variance (disturbance = $-.001$) in the final model, which raised concerns on potential Heywood cases.

Following Chen et al. (2001), we checked model identification, examined model convergence for the model without constraint on residual variance, evaluated the data for extreme response patterns, and compared the models with and without constraint on residual variance. Based on findings, we were able to conclude that the negative residual variance of Self-Monitor was more likely due to sampling variability. Hence, as recommended by Chen et al. (2001), we constrained the residual variance of Self-Monitor to zero in the final model to correct for sampling variability.

Table 5 includes standardized factor loadings on the first- and second-order factors in the model. Loadings of first-order factors (scales) ranged from .69 to .98, signifying strong associations between items and the scales they are designed to measure. Similarly, loadings on the second-order factors (indexes) varied from .67 to .97, indicating strong associations between scales and their respective indexes. Furthermore, the correlation between CRI and B-ERI was .67 ($p < .01$), signifying a large positive correlation between the two indexes.

We calculated Omega coefficients and found strong composite reliability of each second-order factor (CRI = .99, B-ERI = .99) and each first-order factor (Inhibit = .98, Self-Monitor = .99, Shift = .88, Emotional Control = .97, Initiate = .97, Working Memory = .98, Plan/Organize = .97, Task-Monitor = .98, Organization of Materials = .95).

Criterion-Related Validity

Results of the convergent and predictive validity analyses revealed a range of correlations that were in the expected direction. We use Cohen’s (1992) guidelines to interpret the strength of the relationship (.10 = small, .30 = medium, .50 = large). For convergent validity correlation analyses, we found that all BRIEF-2 factors negatively correlated with both the Externalizing and Internalizing scales on the CAB-T, with medium to large effect sizes. Notably, the B-ERI correlated most strongly with the CAB-T Externalizing scores, with coefficients for associated first order factors ranging from $-.45$ (Shift) to $-.72$ (Inhibit). Although correlations between the BRIEF-2 factors and CAB-T Internalizing scores showed more variability, they were most strongly associated with Emotional Control ($-.47$) and Shift ($-.44$). Results of the multiple regression model indicated that both CRI ($\beta = -.14$, $t = -4.98$, $p < .01$) and B-ERI ($\beta = -.52$, $t = -18.18$, $p < .01$) were significant predictors of CAB-T Internalizing scores. In the model for BRIEF-2 second-order factors predicting CAB-T Externalizing scores, B-ERI was a significant predictor ($\beta = -.86$, $t = -45.77$, $p < .01$), but CRI was not significant ($\beta = .02$, $t = -1.08$, $p < .28$). Because higher scores on the BRIEF-2 indicate EF deficits and lower scores on the CAB-T indicate behavioral impairments, the findings suggest that lower individual EF on both BRIEF-2 indexes

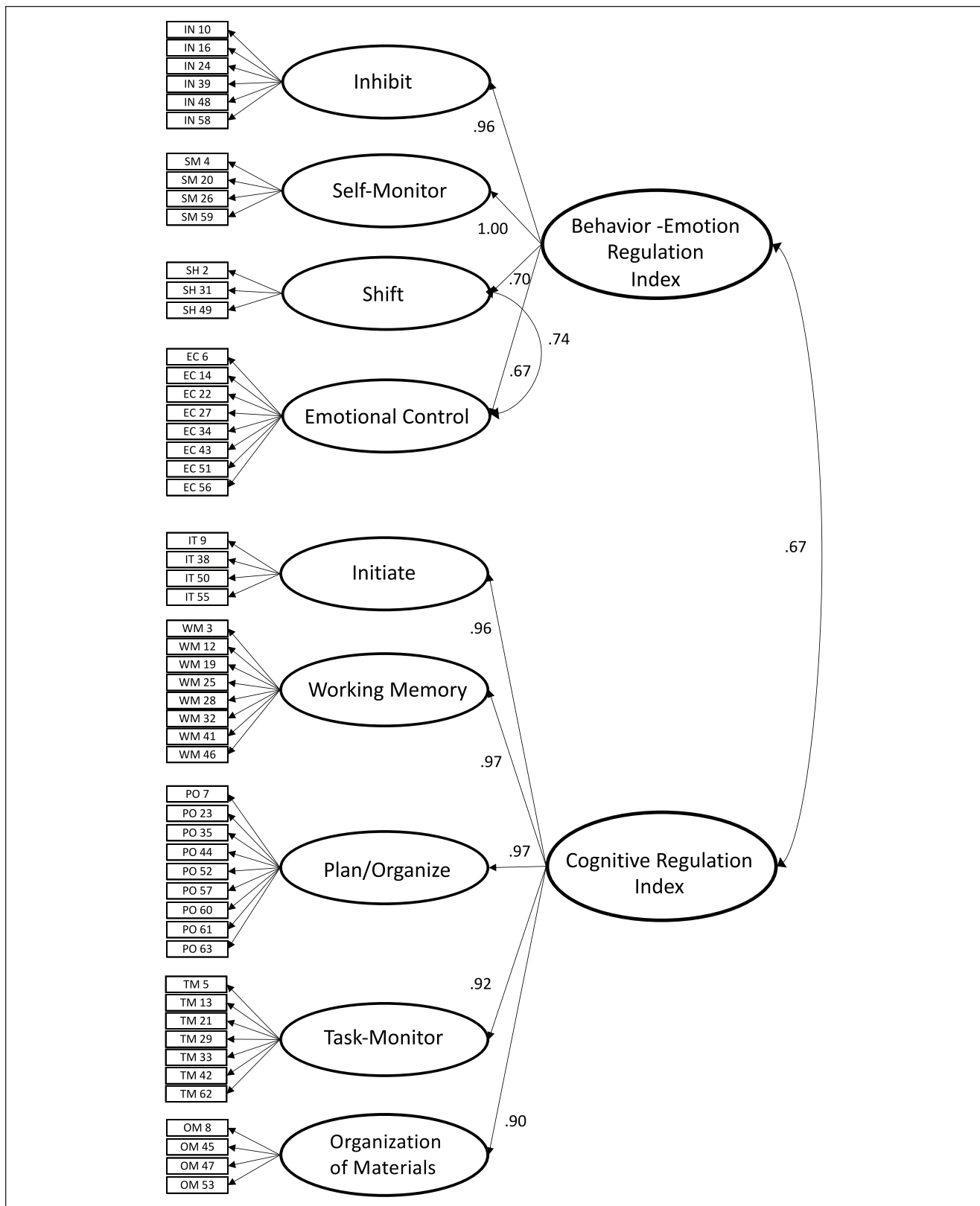


Figure 2. Final BRIEF-2 model.

Note. Standardized path coefficients were presented. PO60 and PO63 loaded on Shift, and TM13 loaded on Self-Monitor in BRIEF–Second edition manual (see SM13, SH60, SH63 in Figure 1). IN = Inhibit; SM = Self-Monitor; EC = Emotional Control; SH = Shift; IT = Initiate; WM = Working Memory; PO = Plan/Organize; TM = Task-Monitor; OM = Organization of Materials.

Table 5. Standardized Factor Loadings From the Final Model.

Model	Loadings (first-order factors)	Loadings (second-order factors)
<i>Behavior–emotion regulation index (B-ERI)</i>		
Inhibit (IN)		.96
IN10	.93	
IN16	.92	
IN24	.87	
IN39	.90	
IN48	.94	
IN58	.98	
Self-Monitor (SM)		1.00
SM04	.91	
SM20	.96	
SM26	.96	
SM59	.96	
Shift		.70
SH02	.95	
SH31	.80	
SH49	.77	
Emotional Control (EC)		.67
EC06	.91	
EC14	.95	
EC22	.90	
EC27	.93	
EC34	.88	
EC43	.86	
EC51	.88	
EC56	.85	
<i>Cognitive regulation index (CRI)</i>		
Initiate (IT)		.96
IT09	.90	
IT38	.92	
IT50	.98	
IT55	.81	
Working Memory (WM)		.97
WM03	.83	
WM12	.91	
WM19	.88	
WM25	.93	
WM28	.85	
WM32	.95	
WM41	.81	
WM46	.93	
Plan/Organize (PO)		.97
PO07	.85	
PO23	.79	
PO35	.69	
PO44	.69	
PO52	.90	
PO57	.92	
PO60 ^a	.85	
PO61	.75	
PO63 ^a	.90	

(continued)

Table 5. (continued)

Model	Loadings (first-order factors)	Loadings (second-order factors)
Task-Monitor (TM)		.92
TM05	.90	
TM13 ^a	.78	
TM21	.93	
TM29	.93	
TM33	.85	
TM42	.93	
TM62	.98	
Organization of Materials (OM)		.90
OM08	.89	
OM45	.85	
OM47	.92	
OM53	.84	

^aIndicate item loaded on different factor than specified in Behavior Rating Inventory of Executive Function–Second edition manual.

was associated with increased maladaptive internalizing behavior, but only B-ERI was related to externalizing difficulties.

For the HTKS, correlations were small but statistically significant across all CRI related scales and ranged from $-.14$ (Organization of Materials) to $-.22$ (Working Memory). In contrast, correlations between the HTKS and B-ERI were considerably smaller with two coefficients (Inhibit, Emotional Control) not reaching statistical significance. The results of the multiple regression model indicated that, controlling for CRI scores, B-ERI was not a significant predictor of HTKS scores ($\beta = -.05$, $t = 1.38$, $p = .17$). On the other hand, controlling for B-ERI scores, CRI was a significant predictor of HTKS scores ($\beta = -.29$, $t = -8.40$, $p < .01$). See Table 6 for all convergent validity results.

Predictive validity results demonstrated a pattern of relationships similar to that of the convergent validity analyses (see Table 7). Specifically, we found significant correlations between both CAB-T Externalizing and Internalizing scores and B-ERI and CRI, with coefficients ranging from small/medium ($-.24$) to large ($-.56$). For the multiple regression model predicting Internalizing scores, results indicated that the CRI ($\beta = -.20$, $t = -4.65$, $p < .01$) and B-ERI ($\beta = -.42$, $t = -9.57$, $p < .01$) were significant predictors. Regarding the model predicting Externalizing scores, B-ERI was a significant predictor ($\beta = -.72$, $t = -19.80$, $p < .01$), whereas CRI was not significant ($\beta = -.01$, $t = -.14$, $p = .89$). These findings suggest that student EF deficits at the beginning of the school year, as indicated on CRI, were related to later internalizing problems, whereas B-ERI scores forecast both internalizing and externalizing behavioral difficulty.

Table 6. Convergent Validity.

Index	CAB-T, INT	CAB-T, EXT	HTKS
Behavior–emotion regulation index	–.374**	–.723**	–.064*
Inhibit	–.193**	–.740**	–.058
Self-Monitor	–.213**	–.669**	–.100**
Shift	–.446**	–.457**	–.074*
Emotional Control	–.471**	–.583**	–.028
Cognitive regulation index	–.222**	–.359**	–.224**
Initiate	–.183**	–.274**	–.202**
Working Memory	–.195**	–.310**	–.229**
Plan/Organize	–.266**	–.357**	–.208**
Task Monitor	–.159**	–.347**	–.206**
Organization of Materials	–.179**	–.322**	–.142**

Note. CAB-T = Clinical Assessment of Behavior–Teacher; EXT = externalizing; INT = internalizing; HTKS = Head-Toes-Knees-Shoulder.

* $p < .05$. ** $p < .01$.

Table 7. Predictive Validity.

Index	CAB-T INT	CAB-T EXT
Behavior–emotion regulation index	–.333**	–.567**
Inhibit	–.195**	–.566**
Self-Monitor	–.225**	–.522**
Shift	–.372**	–.345**
Emotional Control	–.385**	–.476**
Cognitive regulation index	–.249**	–.283**
Initiate	–.181**	–.212**
Working Memory	–.214**	–.234**
Plan/Organize	–.299**	–.291**
Task Monitor	–.197**	–.285**
Organization of Materials	–.209**	–.257**

Note. CAB-T = Clinical Assessment of Behavior–Teacher; EXT = externalizing; INT = internalizing.

* $p < .05$. ** $p < .01$.

Discussion

Given the role EF plays in children's cognitive and psychological development (Blair & Diamond, 2008; Bull et al., 2008; Schoemaker et al., 2012; Zelazo, 2020) and the widespread use of the BRIEF to identify clinically significant behaviors in schools (Roth et al., 2014), the aim of this study was to investigate the factor structure of the BRIEF-2 Teacher Form at the item level with a sample of students in kindergarten and first grade. We were interested in determining whether the underlying theory of the BRIEF-2 was supported by empirical evidence during a period when students experience active EF maturation and are at-risk for developing emotional and behavioral problems. Although research has examined the factor structure of the Parent Form of the BRIEF-2 at the item level (Jacobson et al., 2019), to our knowledge, this is the first study to conduct item level analyses of the teacher version of the BRIEF-2 to

establish internal validity with students in early elementary school at-risk for behavior problems.

Using an iterative EFA and CFA process, we note several important findings. We found that due to inadmissible parameter estimates, there was limited support for the a priori manual model made up of three indexes and nine scales (Gioia et al., 2015), when investigating at the item level. Rather, we discovered overlap between BRI and ERI, indicating the presence of a more general index that includes behavior and emotion regulation. Thus, for this sample of kindergarten and first grade students at risk for developing emotional and behavioral disorders, a modified 53-item (excluding infrequency scale items) second-order model not only fit the data adequately, but resulted in distinct CRI and B-ERI (combined BRI and ERI) factors, which aligns better with the previous theorized model of the original BRIEF, made up of two broad indexes (Gioia et al., 2000).

For students in early elementary school, a more general B-ERI may reflect less of a distinction between BRI and ERI processes at this age. As individuals mature, they develop more sophisticated neurophysiological, cognitive, and language capabilities that effect their regulation and expression of emotions and associated behaviors (Diaz et al., 2017; Gross, 2002). Thus, the degree of unity, independence, and sophistication amongst EF domains may change as children age (Best & Miller, 2010), which, in turn not only affects the extent to which distinct EF processes relate to students' ability to regulate and modulate their behaviors (Riggs et al., 2004; Schoemaker et al., 2013) but also the degree to which teachers are able to report distinct contextualized EF behaviors in the classroom. Thus, for children in kindergarten and first grade at risk for emotional and behavioral disorders, similar processes may be used for both emotion and behavior regulation.

Although an emerging body of research on the BRIEF-2 has found support for three distinct second order factors,

independent studies have tended to include either a wide range of ages (5-18 years old; Jacobson et al., 2019) or did not include students in kindergarten and first grade (8-12 years old; Muñoz & Filippetti, 2019), limiting insight into validity for these students. Furthermore, previous analyses of the original BRIEF resulted in support for a factor structure that did not align with the manual model (Egeland & Fallmyr, 2010; Gioia et al., 2002; Peters et al., 2012), as well as cross loadings of subscales on indexes (Donders et al., 2010; Lyons Usher et al., 2016), highlighting internal validity concerns. Thus, although the developers of the BRIEF-2 (Gioia et al., 2015) normed by age group, these norms may be less reliable due to issues with internal validity. Therefore, our findings provide empirical evidence that for children in kindergarten and first grade at-risk for behavior problems, teachers' perceptions of their contextualized EF skills in the classroom may be distinct from preschool and older elementary school-age students.

We found several items did not load strongly onto the manual specified index or loaded onto more than one construct. For instance, three items (Items 13, 60, and 63) that theoretically align with BRI or ERI, loaded more strongly onto CRI. As these items are phrased in ways that capture more cognitive strategies (i.e., "thinking," "understanding"), for this age group, they may be more reflective of the use of cognitive regulation processes in the classroom (e.g., Task Monitor, Planning/Organizing) rather than the developer theorized Self-Monitoring or Shifting constructs, which may still be developing at this age (Best & Miller, 2010). Furthermore, we found several items had loadings below .40 or exhibited cross loadings, indicating that for our sample, these items likely do not serve as strong EF indicators.

Overall, our item-level findings highlight that while parceling, which was used to establish BRIEF-2 validity (Gioia et al., 2015), can be effective at clarifying multidimensional constructs, they do not permit identification of model misspecifications that stem from item level properties (Little et al., 2013). This is particularly relevant for constructs associated with EF and self-regulation maturation, which are posited to develop throughout childhood and adolescence (Best & Miller, 2010; Huizinga et al., 2006). Thus, our results provide evidence that for students in these early grade levels who are at risk for developing emotional and behavioral problems, 53 items effectively capture contextualized EF.

Results of the criterion-related validity analyses offered further evidence for our proposed BRIEF-2 model both in terms of significant associations with theoretically aligned constructs (convergent validity) and later behavioral outcomes (predictive validity). First, although this is the first known study to examine the relationship between scores on the BRIEF-2 (Gioia et al., 2015) Teacher Form and the HTKS (Ponitz et al., 2009), the small magnitude of the coefficients is consistent with previous findings on the

association between rating scales and performance-based assessments of EF (Toplak et al., 2013). Yet relationships between B-ERI related scales and HTKS scores were either very small or did not reach statistical significance (e.g., Inhibit), which differs with researchers who posit that the HTKS taps primarily inhibitory processes (e.g., Fuhs & Day, 2011). Rather, we found that CRI associated scales, notably Working Memory scale, related most strongly with HTKS scores. Together with the significant regression model results for CRI, this suggests that, among our sample, teacher reports of student's cognitive regulation skills were associated primarily with their performance on the HTKS. Although behavioral risk profiles may play a role, these findings highlight the link between contextualized EF related to cognitive regulation and "cool" EF skills, which aligns with previous studies in which academic achievement (reliant on cognitive skills) has been found to be associated with HTKS scores (Ponitz et al., 2009).

Results of the association between BRIEF-2 factors and scores on the CAB-T indicated a strong relationship between the B-ERI index and associated first order indicators with Externalizing behavioral difficulties. These findings align with the established link between EF deficits and outward-directed problem behaviors among young children (see meta-analysis, Schoemaker et al., 2013). Similarly, scores on B-ERI were related with Internalizing problem behaviors, with the largest coefficients for Emotional Control and Shift. Similarly, Isquith et al. (2014), who examined correlations between the BRIEF-2 and the Internalizing composite scale of the Child Behavior Checklist-Teacher Report Form (Achenbach, 1991), also found a significant link. The association between Shift and teacher reports of internalizing behaviors are particularly important given that items (e.g., Item 31 about being upset) are conceptually related to inward-directed behavioral problems (Gresham & Kern, 2004). Our regression model results, however, indicated that the CRI was not related to externalizing behavior problems. This finding may be due to CRI primarily reflecting children's difficulties with controlling or managing cognitive processes (e.g., short attention span). For instance, several studies have found that young children who demonstrate low attentional control and low negative emotionality (e.g., decreased anger reactivity or emotionally driven aggression) are less likely to demonstrate externalizing behavior (Eisenberg et al., 2000; Morris et al., 2014).

As expected, results of the predictive validity analyses indicated that scores on the B-ERI factors at the beginning of the school year predicted teacher reports of externalizing and internalizing behavior problems approximately 6 months later. These findings are consistent with established research highlighting the predictive role of individual EF and maladaptive behaviors among young children (Lonigan et al., 2017; Morgan et al., 2019). Overall, these results provide further evidence of the BRIEF-2's utility in identifying

students with or at-risk for significant behavioral challenges in early elementary school.

In sum, a modified 53-item, two-factor, second order model made up of general B-ERI and CRI shows a more parsimonious assessment of at-risk kindergarteners and first graders' use of EF skills in the classroom. Because the BRIEF is a well-known tool to distinguish contextualized EF abilities (Alloway et al., 2009; Toplak et al., 2008), the continued use of a three-index model with kindergarten and first grade students may result not only in unreliable findings but also misidentification of students with or at risk for emotional and behavioral problems and associated intervention efforts. Thus, although additional studies are warranted to further establish validation of the current model, school professionals and researchers may consider the clinical utility of the revised BRIEF-2.

Limitations and Future Research

The current study has many notable strengths that include a large demographically diverse sample, the use of an iterative validation process that included both CFA and EFA, and various measures to assess convergent and predictive validity. We also acknowledge several limitations. First, we collected data only from kindergarten and first grade students from one state, which limits the generalizability of our findings to older children or youth as well as students in different locales. Additionally, a large percentage of our sample received free or reduced lunch, which is an indication of low socioeconomic status—a factor related to home, neighborhood, and school stressors (Donovan & Cross, 2002) that are suggested to influence EF development (Cumming et al., 2020; Raver et al., 2013) and related behaviors (Hackman et al., 2015). Accordingly, researchers should continue to evaluate the psychometric properties of the BRIEF-2 with students across all grade levels (e.g., upper elementary, middle, and high school), from different geographic regions, and among children from different socioeconomic backgrounds. Furthermore, because our sample included children who were identified as at risk for emotional and behavioral disorders, additional research will need to be conducted with kindergarten and first graders who are not at risk for behavior problems. Although assessing measurement invariance was beyond the scope of this study, it will be important for future research to assess the measurement invariance of the BRIEF-2, especially related to gender and grade level. Researchers should also examine whether parent rating forms yield similar results due to previous research suggesting low to moderate levels of agreement between parent and teacher reports of EF among children with behavior problems (Dekker et al., 2017; Gioia et al., 2000; McCandless & O'Laughlin, 2007). Last, future EF research would benefit from a shared theoretical understanding of EF, as well as additional insight into which

specific EF constructs performance based and rating scale assess (Isquith et al., 2014). Establishing a shared understanding will support greater consistency in how EF is conceptualized and measured.

Implications for Practice

Our findings have important implications for practice in terms of school-based identification and intervention efforts. In practice, the BRIEF-2 has been shown to be beneficial in early identification of EF difficulties (Isquith et al., 2014; Miranda et al., 2015) and sensitive to clinically significant behaviors (Isquith et al., 2013; Roth et al., 2014). Given the increasing adverse outcomes for students who have early onset emotional and behavioral problems (e.g., Hamre & Pianta, 2001), and the low number of students identified and provided school-based services for mental health problems (Kauffman & Landrum, 2017; Office of Special Education Programs, 2017), a modified version of BRIEF-2 can help school professionals identify at-risk kindergarten and first grade students in a timely manner.

In turn, BRIEF-2 findings can inform the provision of school-based services for students with or at risk for emotional and behavioral disorders. As stipulated by Isquith et al. (2014), to fully capture student EF functioning school professionals should use data from both ratings scales (i.e., modified BRIEF-2) and performance-based EF assessments (e.g., NIH Toolbox; Weintraub et al., 2013) to identify patterns of EF dysfunction and inform targeted prevention and intervention efforts. For instance, findings can (a) aid in the development of context-specific, real-world interventions (Isquith et al., 2014); (b) be embedded within individualized education programs (i.e., special education programming) and 504 plans (depending on the degree of individual need; Gioia & Isquith, 2002); and (c) determine program effectiveness through progress monitoring.

Fortunately, a growing body of research has underscored the effectiveness of interventions that target real-world manifestations of EF in school settings including SELF (Daunic et al., 2013), Incredible Years (Webster-Stratton & Reid, 2004), and the preschool Promoting Alternative Thinking Strategies curriculum (Domitrovich et al., 2007). These programs have shown to be effective at improving young children's self-regulation skills necessary for behavioral and social-emotional competence in the classroom, as well as school readiness. Furthermore, a growing body of research suggests that programming that explicitly targets EF has shown promising results at enhancing skills across age groups (e.g., Tools of the Mind [Blair & Raver, 2014], I Control [Smith et al., 2017]). Overall, using valid and reliable EF rating scales during the transition to elementary school is a critical component of accurately assessing factors underlying psychological functioning (Goldstein & Flake, 2016) and informing the

provision of services to mitigate the potential for adverse mental health problems in young children (Bradley et al., 2008; Kauffman & Landrum, 2017).

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